

Partial Dependence Plots

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Introduction

Partial dependence plots (PDPs), introduced by Friedman in 2001, show the average model prediction as a feature varies while holding other features at their observed values. They provide a visual answer to “how does the model’s prediction change with x_j ?” — exposing non-linear effects and interactions in black-box models that variable-importance scores alone cannot reveal. Combined with variable importance (which features matter) and SHAP values (per-prediction explanations), PDPs are the third pillar of interpretation for opaque ML models.

Prerequisites

A working understanding of trained black-box classifiers or regressors (random forest, gradient boosting, neural networks), variable importance, and the difference between marginal and conditional effects.

Theory

Partial dependence at value $x_j = v$ is

$$\text{PD}_j(v) = \frac{1}{n} \sum_{i=1}^n \hat{f}(v, x_{-j,i}),$$

i.e., set the feature of interest to v across all rows in the dataset, predict, and average. The PDP is the curve of $\text{PD}_j(v)$ as v varies over the observed range of x_j .

Individual Conditional Expectation (ICE) plots show one curve per observation rather than the average; the PDP is the mean of the ICE curves. ICE plots expose heterogeneity in feature effects: if all ICE curves are roughly parallel, the effect of x_j is homogeneous; if they diverge, an interaction with another feature is implied.

Accumulated Local Effects (ALE) plots address PDP’s extrapolation problem when features are correlated; they integrate the prediction’s local derivative rather than averaging predictions at potentially out-of-distribution feature combinations.

Assumptions

For PDP interpretation, features are approximately independent; otherwise the average over x_{-j} at fixed $x_j = v$ may include low-density regions of the joint distribution and produce misleading curves. For correlated features, use ALE plots instead.

R Implementation

```
library(pdp); library(ranger)

set.seed(2026)
n <- 500
```

```
df <- data.frame(
  x1 = rnorm(n), x2 = rnorm(n), x3 = rnorm(n),
  y = sin(rnorm(n)) + 0.3 * rnorm(n)^2
)

rf <- ranger(y ~ ., data = df, num.trees = 300)
pd <- partial(rf, pred.var = "x1", plot = FALSE, train = df)

plot(pd$x1, pd$yhat, type = "l", lwd = 2, col = "#2A9D8F",
     xlab = "x1", ylab = "PD(x1)",
     main = "Partial dependence of x1")
```

Output & Results

`pdp::partial()` returns a data frame with the feature value and the partial-dependence prediction. Plotting it as a curve shows how the average prediction varies with the feature; non-linear shapes reveal effects that linear models would miss.

Interpretation

A reporting sentence: “The partial-dependence plot revealed a non-monotone sinusoidal relationship between x_1 and the outcome (peak prediction near $x_1 = -0.5$, trough near $x_1 = 1.5$); this structure was not visible from variable importance alone, which simply ranked x_1 as the strongest predictor. ICE curves were broadly parallel, indicating no major interaction with other features.” Always pair PDP with ICE for heterogeneity detection.

Practical Tips

- Pair PDP with variable importance: importance answers “which features matter?”, PDP answers “how do they affect the prediction?”.
- Use ICE plots to detect interactions: ICE curves diverging from the PDP indicate heterogeneous effects driven by another feature.
- PDPs extrapolate into low-density regions when features are correlated; for correlated-feature contexts, use Accumulated Local Effects (ALE) plots via `iml::FeatureEffect` or `ALEPlot`.
- Two-dimensional PDPs (pair of features) reveal interactions but are computationally expensive; use a sampled feature grid for large datasets.
- `DALEX::model_profile()` offers a unified interpretation interface across tree, linear, and deep models with consistent API.
- For per-prediction explanations rather than global ones, use SHAP values (`shapviz`, `iml`); SHAP and PDP answer related but distinct interpretation questions.

R Packages Used

`pdp` for the canonical PDP implementation; `iml::FeatureEffect()` for PDP, ICE, and ALE within a unified interpretation framework; `DALEX::model_profile()` for model-agnostic PDPs; `ALEPlot` for ALE specifically; `shapviz` and `fastshap` for SHAP-based per-prediction explanations.

For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren’t.
- Red flags in tables and figures.

- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Cross-validation, nested CV, bootstrap .632+
- Regularisation: ridge, lasso, elastic net
- Trees, random forests, gradient boosting
- Interpretability and SHAP
- Tabular neural networks with torch
- Imaging and sequence models intro
- tidymodels pipelines
- FDR, knockoffs, replication crisis